1. Detailed Calculations for Inferring HMM-stabilized Deep Learning.

1.1. *Parameter Estimation via the EM Algorithm.* In principle, the parameters of HMM-stabilized deep learning model can be estimated via the maximum likelihood principle, which is a function of both true and predicted tool labels with respect to $\boldsymbol{\theta}$. Because, true phase and tool labels are observed for training data only, we typically rely on the expectation-maximization algorithm Dempster, Laird and Rubin (1977) to do the optimization, treating the unobserved tool labels as missing data. The Q-function of E-step is shown as follows:

$$Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(s)}) = \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \log \mathbb{P}(\mathcal{I}_i) \mathbb{P}\left(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}\right), \tag{1}$$

where $\boldsymbol{\theta}^{(s)}$ is the estimation of model parameter $\boldsymbol{\theta}$ in the s-th iteration of the EM algorithm.

After substituting the value of $\mathbb{P}(\mathcal{I}_i)$ into the Q-function, we obtain

$$\begin{aligned}
Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(s)}) = & \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \log \left(\textbf{Multinomial}(P_{i,1}|\boldsymbol{\alpha}) \cdot \prod_{t=2}^{n_i} \mathbf{A}(P_{i,t-1}, P_{i,t}) \right) \mathbb{P}\left(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}\right) \\
& + \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \log \left(\prod_{\tau \in \mathcal{T}} \textbf{Bernoulli}(T_{i,1,\tau}|\beta_{\tau,P_{i,1}}) \cdot \prod_{t=2}^{n_i} \mathbf{A}_{\tau,P_{i,t}}(T_{i,t-1,\tau}, T_{i,t,\tau}) \right) \mathbb{P}\left(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}\right) \\
& + \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \log \left(\prod_{t=1}^{n_i} \mathbf{B}(P_{i,t}, \hat{P}_{i,t}) \right) \mathbb{P}\left(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}\right) \\
& + \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \log \left(\prod_{\tau \in \mathcal{T}} \prod_{t=1}^{n_i} \mathbf{B}_{\tau}(T_{i,t,\tau}, \hat{T}_{i,t,\tau}) \right) \mathbb{P}\left(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}\right).
\end{aligned} \tag{2}$$

By further grouping similar terms in Eq. (2) based on the parameters $\boldsymbol{\theta} = (\boldsymbol{\alpha}, \mathbf{A}; \boldsymbol{\beta}, \mathcal{A}; \mathbf{B}, \mathcal{B})$, we have

$$\begin{aligned}
Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(s)}) = & \sum_{\varrho \in \mathcal{P}} \log \alpha_{\varrho} \mathbb{E}\left[\mathbb{N}(P_{\cdot,1} = \varrho | \boldsymbol{\theta}^{(s)})\right] \\
& + \sum_{\varrho_i \in \mathcal{P}} \sum_{\varrho_j \in \mathcal{P}} \log \mathbf{A}(\varrho_i, \varrho_j) \mathbb{E}\left[\mathbb{N}(P_{\cdot,t-1} = \varrho_i, P_{\cdot,t} = \varrho_j | \boldsymbol{\theta}^{(s)})\right] \\
& + \sum_{\tau \in \mathcal{T}} \sum_{l=0}^{1} ((1-l)\log(1-\beta_{\tau,\varrho}) + l \log(\beta_{\tau,\varrho})) \mathbb{E}\left[\mathbb{N}(T_{\cdot,1,\tau} = l, P_{\cdot,1} = \varrho | \boldsymbol{\theta}^{(s)})\right] \\
& + \sum_{\tau \in \mathcal{T}} \sum_{\varrho \in \mathcal{P}} \sum_{m=0}^{1} \sum_{n=0}^{1} \log \mathbf{A}_{\tau,\varrho}(m,n) \mathbb{E}\left[\mathbb{N}(T_{\cdot,t-1,\tau} = m, T_{\cdot,t,\tau} = n, P_{\cdot,t} = \varrho | \boldsymbol{\theta}^{(s)})\right] \} \\
& + \sum_{\varrho_i \in \mathcal{P}} \sum_{\varrho_j \in \mathcal{P}} \log \mathbf{B}(\varrho_i, \varrho_j) \mathbb{E}\left[\mathbb{N}(P_{\cdot,t} = \varrho_i, \hat{P}_{\cdot,t} = \varrho_j | \boldsymbol{\theta}^{(s)})\right] \\
& + \sum_{\tau \in \mathcal{T}} \sum_{m=0}^{1} \sum_{n=0}^{1} \log \mathbf{B}_{\tau}(m,n) \mathbb{E}\left[\mathbb{N}(T_{\cdot,t,\tau} = m, \hat{T}_{\cdot,t,\tau} = n | \boldsymbol{\theta}^{(s)})\right],
\end{aligned} \tag{3}$$

where

$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(P_{i,1}=\varrho)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,1,\tau}=j,P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(T_{i,1,\tau}=j,P_{i,1}=\varrho)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,t-1}=\varrho_i,P_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(P_{i,t-1}=\varrho_i,P_{i,t}=\varrho_j)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,t}=\varrho_i,\hat{P}_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(P_{i,t}=\varrho_i,\hat{P}_{i,t}=\varrho_j)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t-1,\tau}=i,T_{\cdot,t,\tau}=j,P_{\cdot,t}=\varrho|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(T_{i,t-1,\tau}=i,T_{i,t,\tau}=j,P_{i,t}=\varrho)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t,\tau}=i,\hat{T}_{\cdot,t,\tau}=j|\boldsymbol{\theta}^{(s)})\right]=\sum_{i=1}^{m}\sum_{\mathcal{I}_i}\mathbb{I}(T_{i,t,\tau}=i,\hat{T}_{i,t,\tau}=j)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}).$$

To identify the value of $\boldsymbol{\theta}$ that maximizes the Q-function, we solve the equation by setting the derivatives of the Q-function with respect to $\boldsymbol{\theta}$ to zero in M-step. Considering $\sum_{\varrho\in\mathcal{P}}\alpha_\varrho=1$, we apply the Lagrange multiplier method to determine the optimal value of $\{\alpha_\varrho\}_{\varrho\in\mathcal{P}}$, resulting in the following equation,

$$\frac{\partial Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(s)})-\lambda(\sum_{\varrho\in\mathcal{P}}\alpha_\varrho-1)}{\partial\alpha_\varrho}=0,(\varrho\in\mathcal{P}). \tag{4}$$

Eq. (4) is simplified as follows,

$$\frac{\mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right]}{\alpha_\varrho}-\lambda=0,(\varrho\in\mathcal{P}). \tag{5}$$

After taking the sum of Eq. (5) over all $\varrho\in\mathcal{P}$, we have

$$\sum_{\varrho\in\mathcal{P}}\mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right]=\sum_{\varrho\in\mathcal{P}}\lambda\alpha_\varrho=\lambda. \tag{6}$$

When we substitute Eq. (6) into Eq. (5), we derive the M-step updating function for the parameters $\boldsymbol{\alpha}=\{\alpha_\varrho\}_{\varrho\in\mathcal{P}}$. By applying the similar Lagrange multiplier method to these parameters, we can also derive the updating function of other parameters in the M-step.

This estimation process can be implemented using a standard Baum-Welch algorithm. We employ the standard forward-backward algorithm to compute the six expectations mentioned in Eq. (3), given the complexity of enumerating hidden states (refer to Appendix 1.2 for more details).

1.2. *Fast Computation of the E-Step.* In the parameter estimation steps of HMM-stabilized methods, we use forward-backward algorithm to get the expectation of interest in E-step. Due to the complexity of hidden state enumeration, we utilize the standard forward-backward algorithm to calculate the expectation in Eq. (3) via EM algorithm.

For the observation $(\hat{\mathbf{P}}_i, \hat{\mathbf{T}}_i)$ of n_i key frames in video i and parameters $\boldsymbol{\theta}^{(t)}$ of the HMM-stabilized model, we define the forward and backward variables as follows,

$$\mathbb{U}_{i,t}(\varrho,\tau) = \mathbb{P}(\hat{\mathbf{P}}_{i,[n\le t]}, \hat{\mathbf{T}}_{i,[n\le t]}, P_{i,t}=\varrho, T_{i,t,1}=\tau_1,\cdots,T_{i,t,K}=\tau_K), \tag{7}$$

$$\mathbb{V}_{i,t}(\varrho,\tau) = \mathbb{P}(\hat{\mathbf{P}}_{i,[n> t]}, \hat{\mathbf{T}}_{i,[n> t]} | P_{i,t}=\varrho, T_{i,t,1}=\tau_1,\cdots,T_{i,t,K}=\tau_K) \quad (1\le t\le n_i). \tag{8}$$

where

$$\tau=(\tau_1,\cdots,\tau_K), \hat{\mathbf{P}}_{i,[n\le t]} = (\hat{P}_{i,1},\cdots,\hat{P}_{i,t}), \hat{\mathbf{P}}_{i,[n>t]} = (\hat{P}_{i,t+1},\cdots,\hat{P}_{i,n_i}),$$

$$\hat{\mathbf{T}}_{i,[n\le t]} = (\hat{T}_{i,1},\cdots,\hat{T}_{i,t}), \hat{\mathbf{T}}_{i,[n>t]} = (\hat{T}_{i,t+1},\cdots,\hat{T}_{i,n_i}).$$

The forward and backward variables can be computed using the following dynamic programming iteration formula,

$$\mathbb{U}_{i,1}(\varrho,\tau) = \mathbb{P}(P_{i,1}=\varrho)\mathbb{P}(\hat{P}_{i,1}|P_{i,1}=\varrho)\prod_{k=1}^{K}\mathbb{P}(T_{i,1,k}=\tau_k)\mathbb{P}(\hat{T}_{i,1,k}|T_{i,1,k}=\tau_k), \tag{9}$$

$$\begin{aligned}\mathbb{U}_{i,t+1}(\varrho,\tau) = &\sum_{\varrho^*=1}^{L}\sum_{\tau_1^*=0}^{1}\cdots\sum_{\tau_K^*=0}^{1}\mathbb{U}_{i,t}(\varrho^*,\tau^*)\mathbb{P}(P_{i,t+1}=\varrho|P_{i,t}=\varrho^*)\mathbb{P}(\hat{P}_{i,t+1}|P_{i,t+1}=\varrho)\\ &\prod_{k=1}^{K}\mathbb{P}(T_{i,t+1,k}=\tau_k|T_{i,t,k}=\tau_k^*)\mathbb{P}(\hat{T}_{i,t+1,k}|T_{i,t+1,k}=\tau_k);\end{aligned} \tag{10}$$

(11)

$$\begin{aligned}\mathbb{V}_{i,n_i}(\varrho,\tau) = &1,\\ \mathbb{V}_{i,t-1}(\varrho,\tau) = &\sum_{\varrho^*=1}^{L}\sum_{\tau_1^*=0}^{1}\cdots\sum_{\tau_K^*=0}^{1}\mathbb{V}_{i,t}(\varrho^*,\tau^*)\mathbb{P}(P_{i,t}=\varrho^*|P_{i,t-1}=\varrho)\mathbb{P}(\hat{P}_{i,t-1}|P_{i,t-1}=\varrho)\\ &\prod_{k=1}^{K}\mathbb{P}(T_{i,t-1,k}=\tau_k^*|T_{i,t,k}=\tau_k)\mathbb{P}(\hat{T}_{i,t-1,k}|T_{i,t-1,k}=\tau_k).\end{aligned} \tag{12}$$

Note that

$$\mathbb{P}(\hat{\mathbf{T}},\hat{\mathbf{P}}|\boldsymbol{\theta}^{(t)}) = \sum_{\varrho\in\mathcal{P}}\sum_{\tau\in\mathcal{T}}\sum_{i=1}^{m}\mathbb{U}_{i,n_i}(\varrho,\tau). \tag{13}$$

The expectations can be computed using the following formulas,

(14)

$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho)|\boldsymbol{\theta}^{(t)}\right] = \sum_{i=1}^{m}\sum_{\tau\in\mathcal{T}}\mathbb{U}_{i,1}(\varrho,\tau)\mathbb{V}_{i,1}(\varrho,\tau)\Big/\mathbb{P}(\hat{\mathbf{T}},\hat{\mathbf{P}}|\boldsymbol{\theta}^{(t)}),$$

$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,t-1}=\varrho,P_{\cdot,t}=\varrho^*)|\boldsymbol{\theta}^{(t)}\right] =$$

$$\sum_{i=1}^{m}\sum_{\tau\in\mathcal{T}}\sum_{\tau^*\in\mathcal{T}}\sum_{t=1}^{n_i-1}\mathbb{U}_{i,t}(\varrho,\tau)\mathbb{V}_{i,t+1}(\varrho^*,\tau^*)\mathbb{P}(P_{i,t+1}=\varrho^*|P_{i,t}=\varrho)\mathbb{P}(\hat{P}_{i,t+1}|P_{i,t+1}=\varrho^*)$$

(15)
$$\times \prod_{k=1}^{K} \mathbb{P}(T_{i,t,k} = \tau_k | T_{i,t+1,k} = \tau_k^*) \mathbb{P}(\hat{T}_{i,t+1,k} | T_{i,t+1,k} = \tau_k^*) \Big/ \mathbb{P}(\hat{\mathbf{T}}, \hat{\mathbf{P}} | \boldsymbol{\theta}^{(t)}),$$

(16)
$$\mathbb{E}\left[\mathbb{N}(P_{\cdot,t} = \varrho, \hat{P}_{\cdot,t} = \varrho^*) | \boldsymbol{\theta}^{(t)}\right] = \sum_{\tau \in \mathcal{T}} \sum_{\tau^* \in \mathcal{T}} \sum_{(\hat{P}_{i,t}, \hat{T}_{i,t}) = (\varrho^*, \tau^*)} \mathbb{U}_{i,t}(\varrho, \tau) \mathbb{V}_{i,t}(\varrho, \tau) \Big/ \mathbb{P}(\hat{\mathbf{T}}, \hat{\mathbf{P}} | \boldsymbol{\theta}^{(t)}),$$

(17)
$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,1,k} = \tau) | \boldsymbol{\theta}^{(t)}\right] = \sum_{\varrho \in \mathcal{P}} \mathbb{U}_{i,1}(\varrho, \tau) \mathbb{V}_{i,1}(\varrho, \tau) \Big/ \mathbb{P}(\hat{\mathbf{T}}, \hat{\mathbf{P}} | \boldsymbol{\theta}^{(t)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t-1,k} = j, T_{\cdot,t,k} = \tau^*, P_{\cdot,t} = \varrho) | \boldsymbol{\theta}^{(t)}\right] =$$

$$\sum_{\varrho^* \in \mathcal{P}} \sum_{t=1}^{n_i - 1} \mathbb{U}_{i,t}(\varrho, \tau) \mathbb{V}_{i,t+1}(\varrho^*, \tau^*) \mathbb{P}(P_{i,t+1} = \varrho^* | P_{i,t} = \varrho) \mathbb{P}(\hat{P}_{i,t+1} | P_{i,t+1} = \varrho^*)$$

(18)
$$\times \prod_{k=1}^{K} \mathbb{P}(T_{i,t,k} = \tau_k | T_{i,t+1,k} = \tau_k^*) \mathbb{P}(\hat{T}_{i,t+1,k} | T_{i,t+1,k} = \tau_k^*) \Big/ \mathbb{P}(\hat{\mathbf{T}}, \hat{\mathbf{P}} | \boldsymbol{\theta}^{(t)}),$$

(19)
$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t,k} = \tau, \hat{T}_{\cdot,t,k} = \tau^*) | \boldsymbol{\theta}^{(t)}\right] = \sum_{\varrho \in \mathcal{P}} \sum_{\varrho^* \in \mathcal{P}} \sum_{(\hat{P}_{i,t}, \hat{T}_{i,t}) = (\varrho^*, \tau^*)} \mathbb{U}_{i,t}(\varrho, \tau) \mathbb{V}_{i,t}(\varrho, \tau) \Big/ \mathbb{P}(\hat{\mathbf{T}}, \hat{\mathbf{P}} | \boldsymbol{\theta}^{(t)}).$$

1.3. *The Degenerated Cases.* When only tool recognition is considered, we get the degenerated likelihood with parameters $\boldsymbol{\theta} = (\boldsymbol{\beta}, \mathcal{A}, \mathcal{B})$ as the degenerated parameters, resulting in the following Q-function,

$$\begin{aligned} Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) &= \sum_{\tau \in \mathcal{T}} \sum_{j=0}^{1} ((1-j) \log(1 - \beta_\tau) + j \log(\beta_\tau)) \mathbb{E}\left[\mathbb{N}(T_{\cdot,1,\tau} = j | \boldsymbol{\theta}^{(s)})\right] \\ &+ \sum_{\tau \in \mathcal{T}} \sum_{i=0}^{1} \sum_{j=0}^{1} \log \mathbf{A}_\tau(i,j) \mathbb{E}\left[\mathbb{N}(T_{\cdot,t-1,\tau} = i, T_{\cdot,t,\tau} = j | \boldsymbol{\theta}^{(s)})\right] \\ &+ \sum_{\tau \in \mathcal{T}} \sum_{i=0}^{1} \sum_{j=0}^{1} \log \mathbf{B}_\tau(i,j) \mathbb{E}\left[\mathbb{N}(T_{\cdot,t,\tau} = i, \hat{T}_{\cdot,t,\tau} = j | \boldsymbol{\theta}^{(s)})\right], \end{aligned} \tag{20}$$

where $\boldsymbol{\theta}^{(s)}$ is the parameter estimation in the s-th iteration of the EM algorithm, and

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,1,\tau} = j | \boldsymbol{\theta}^{(s)})\right] = \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \mathbb{I}(T_{i,1,\tau} = j) \mathbb{P}(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t-1,\tau} = i, T_{\cdot,t,\tau} = j | \boldsymbol{\theta}^{(s)})\right] = \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \mathbb{I}(T_{i,t-1,\tau} = i, T_{i,t,\tau} = j) \mathbb{P}(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}),$$

$$\mathbb{E}\left[\mathbb{N}(T_{\cdot,t,\tau} = i, \hat{T}_{\cdot,t,\tau} = j | \boldsymbol{\theta}^{(s)})\right] = \sum_{i=1}^{m} \sum_{\mathcal{I}_i} \mathbb{I}(T_{i,t,\tau} = i, \hat{T}_{i,t,\tau} = j) \mathbb{P}(\mathcal{I}_i | \mathcal{I}_i^{obs}, \boldsymbol{\theta}^{(s)}).$$

Similarly, when only phase recognition is considered, we have

$$
\begin{aligned}
Q(\boldsymbol{\theta}|\boldsymbol{\theta}^{(s)}) = &\sum_{\varrho\in\mathcal{P}} \log \alpha_{\varrho} \mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right] \\
&+ \sum_{\varrho_i\in\mathcal{P}} \sum_{\varrho_j\in\mathcal{P}} \log \mathbf{A}(\varrho_i,\varrho_j) \mathbb{E}\left[\mathbb{N}(P_{\cdot,t-1}=\varrho_i, P_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right] \\
&+ \sum_{\varrho_i\in\mathcal{P}} \sum_{\varrho_j\in\mathcal{P}} \log \mathbf{B}(\varrho_i,\varrho_j) \mathbb{E}\left[\mathbb{N}(P_{\cdot,t}=\varrho_i, \hat{P}_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right],
\end{aligned} \tag{21}
$$

where

$$
\begin{aligned}
\mathbb{E}\left[\mathbb{N}(P_{\cdot,1}=\varrho|\boldsymbol{\theta}^{(s)})\right] &= \sum_{i=1}^{m}\sum_{\mathcal{I}_i} \mathbb{I}(P_{i,1}=\varrho)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}), \\
\mathbb{E}\left[\mathbb{N}(P_{\cdot,t-1}=\varrho_i, P_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right] &= \sum_{i=1}^{m}\sum_{\mathcal{I}_i} \mathbb{I}(P_{i,t-1}=\varrho_i, P_{i,t}=\varrho_j)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}), \\
\mathbb{E}\left[\mathbb{N}(P_{\cdot,t}=\varrho_i, \hat{P}_{\cdot,t}=\varrho_j|\boldsymbol{\theta}^{(s)})\right] &= \sum_{i=1}^{m}\sum_{\mathcal{I}_i} \mathbb{I}(P_{i,t}=\varrho_i, \hat{P}_{i,t}=\varrho_j)\mathbb{P}(\mathcal{I}_i|\mathcal{I}_i^{obs},\boldsymbol{\theta}^{(s)}).
\end{aligned}
$$

By following the same procedure outlined in Section 1.1, we can derive the iteration formula of the EM algorithm for the degenerate case. The E-step in the degenerate model can be computed quickly using a similar forward-backward procedure as described in Section 1.2.

REFERENCES

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